Homework Assignment 1

Math 292, Spring 2026

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Problem 1:

(a)

$$\begin{aligned} E[(3+2Y)^2] &= E[9+12Y+4Y^2] \\ &= 9+12E[Y]+4E[Y^2] \\ &= 9+12(2)+4(Var(Y)+E[Y]^2) \\ &= 65 \end{aligned}$$

(b)

$$Var(5+4Y) = 4^2 Var(Y) = 64$$

Problem 2:

(a)

$$P(X=1) = p(HT)+p(TH) = 0.6*0.3+0.4*0.7 = 0.46$$

(b)

$$\begin{aligned} E[X] &= 0*P(X=0)+1*P(X=1)+2*P(X=2) \\ &= 0.46+2*0.42 \\ &= 1.3 \end{aligned}$$

Problem 3:

(a)

$$\begin{aligned} &p(\text{Strategy 1}) = p \\ &p(\text{Strategy 2}) = p^2+\frac{1}{2}*2p(1-p) = p \end{aligned}$$

(b) Strategy 2 has a equal probability as Strategy 1.

(c)

$$E[X] = 10 * 0.7 = 7$$

Problem 4:

(a) Let A be the event that dice A is rolled, and B the event that dice B is rolled and G be the event that green is rolled

$$\begin{aligned} P(G) &= P(G \mid A)P(A) + P(G \mid B)P(B) \\ &= \frac{1}{2} * \frac{1}{2} + \frac{1}{6} * \frac{1}{2} \\ &= \frac{1}{3} \end{aligned}$$

(b) The probability of rolling two greens in a row is

$$\begin{aligned} P(G_1, G_2) &= P(G_1, G_2 \mid A)P(A) + P(G_1, G_2 \mid B)P(B) \\ &= (\frac{1}{2})^3 + (\frac{1}{6})^2(\frac{1}{2}) \\ &= \frac{5}{36} \end{aligned}$$

Then we have that

$$\begin{aligned} P(G_3 \mid G_1, G_2) &= \frac{P(G_1, G_2, G_3 \mid A)P(A) + P(G_1, G_2, G_3 \mid B)P(B)}{P(G_1, G_2)} \\ &= \frac{(\frac{1}{2})^4 + (\frac{1}{6})^3(\frac{1}{2})}{\frac{5}{36}} \\ &= \frac{7}{15} \end{aligned}$$

(c)

$$\begin{aligned} P(A \mid G_1, G_2) &= \frac{P(G_1, G_2 \mid A)P(A)}{P(G_1, G_2)} \\ &= \frac{(\frac{1}{2})^3}{\frac{5}{36}} \\ &= \frac{9}{10} \end{aligned}$$

Problem 5:

Let W be the event of winning \$1.10 and it's complement losing \$1.00

Then, we can solve for the probability of two of the same color marbles being picked

$$P(W) = \frac{\binom{5}{2} + \binom{5}{2}}{\binom{10}{2}} = \frac{4}{9}$$
$$P(W^c) = 1 - P(W) = \frac{5}{9}$$

Therefore, the expected value is just

$$E[X] = 1.1 * P(W) + -1 * P(W^c) \approx -0.067$$

Then, the variance is

$$Var[X] = E[X^2] - E[X]^2 \approx 1.0889$$

Problem 6: Let C_i be the door with the car and M_i the doors monty opens. Lets say we choose door 1, then

$$P(C_1) = \frac{1}{7}$$

for staying.

We choose door 2 as the door we want to switch to, and Monty chooses 3,4,5.

$$\begin{aligned} P(C_2 \mid M_3, M_4, M_5) &= \frac{P(M_3, M_4, M_5 \mid C_2)P(C_2)}{P(M_3, M_4, M_5)} \\ &= \frac{\frac{1}{\binom{5}{3}} * \frac{1}{7}}{\frac{1}{\binom{6}{3}}} = \frac{\frac{1}{10} * \frac{1}{7}}{\frac{1}{20}} \\ &= \frac{2}{7} \end{aligned}$$

Yes, we should switch doors, since the probability of winning when switching is $\frac{2}{7}$ while staying is only $\frac{1}{7}$.